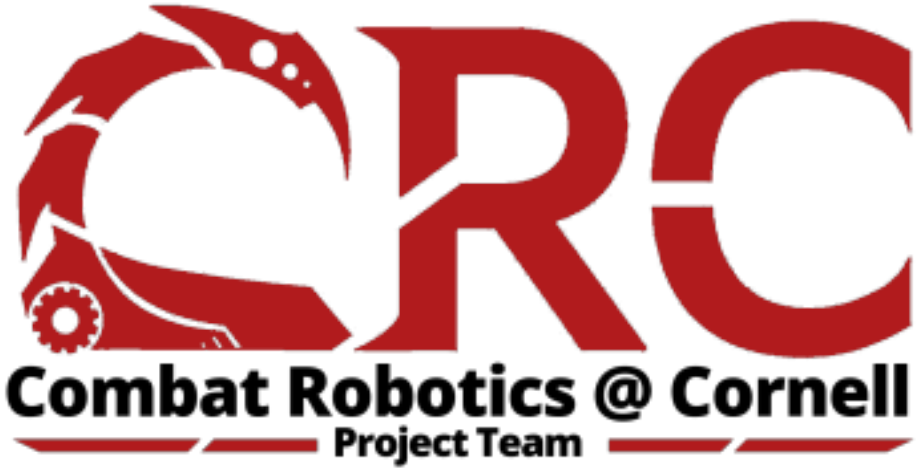


Final Report

Infinity Subteam

California Roll

Spring 2026

Cornell
University

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1 Introduction

1.1 Subteam Purpose

On behalf of the Infinity subteam, our mission was to build a combined total of 5 antweight bots that would work collectively to serve as "The Four Horsemen" in the 3lb category at the May NHRL competition. This team consisted of a variety of different types of antweight bots: two hammersaws and three vertical spinners.

Team Cali, one of the bots within the Four Horsemen, served to design and manufacture a 2WD skid-steer vertical eggbeater spinner. In comparison to team Galvanizer and Stingray, California Roll opted for a stronger and higher kinetic energy weapon system that would intend to hit opponents with larger hits rather than the sawblades that the other teams had. Our primary technical goals this semester were to design a simple high-kinetic-energy weapon and embrace design for manufacturing and assembly (DFMA) in each aspect of our design.

1.2 Subteam Members

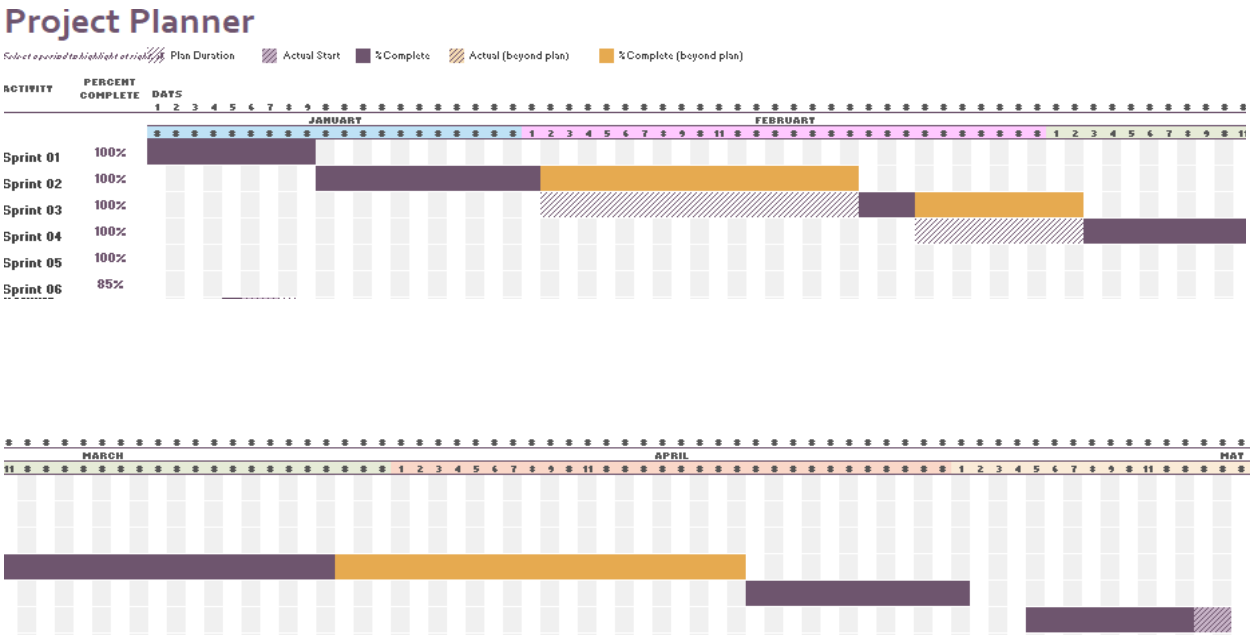
Gavin Lim	Computer Science '29	Infinity Subteam Member
Georgia Tani Green,	Mechanical Engineering '29	Infinity Subteam Member

2 Timeline

The following section details our timeline planning for the semester, along with a discussion of challenges encountered and the measures implemented to resolve them and keep the project on track.

2.1 Timeline Sprints

Across the semester, our design sprints stayed on schedule, driven by focused out-of-meeting effort and disciplined take-home tasks. We delivered a near-final CAD in time for the Infinity Final Design Review, finalized our CAD, assembled and tested our bot in time for the CRC Showcase. Then, we tested our bots some more and prepared for the NHRL May competition. We completed the Spring 2026 report during the exam study window.



Sprint #	Sprint Names	Intended Dates	Actual Dates
Sprint 1	Ideation	1/12-1/20	1/12-1/20
Sprint 2	Subsystem Details and Rough CAD	1/22-2/1	1/22-2/19
Sprint 3 (Final Design Review)	Final CAD	2/2-2/22	2/20-3/3
Sprint 4 (Infinity Competition)	Assembly and Testing	2/23-3/29	3/4 - 4/20
Sprint 5 (prep for NHRL)	Prep for NHRL	4/21-5/2	4/21-5/2
Sprint 6 (write report)	Write report	5/5-5/16	5/5-5/16

2.2 Timeline Reflection

In this section, we reflect on the challenges presented by attempting to make a realistic timeline with no prior experience. This was the first year the team ever attempted a 1lb competition, compressing the traditional Fall/Spring Design/Manufacturing cycle into a one-semester continuous sprint. Additionally, none of us were experienced with any part of the cycle, so how we allocated our time compared to our timeline reflects that this was a new experience for everyone involved.

We had three design reviews this semester, where we convened with more experienced members from other subteams to reflect on and iterate our design. At the first design review (at the end of sprint 1 in the timeline), we brought initial sketches of our bot and the preliminary design of our weapon. For the second design review, we specced our weapon motor and ESC and refined the weapon CAD. Our main goals for that design review were to ensure our weapon would be machinable in-house.

In preparation for our final design review, the date of which was pushed back to accommodate our and other groups' taking longer than anticipated to finalize our CAD and BoM. We utilized a substantial amount of outside-of-meeting work to ensure we had a finalized design ready to present and receive feedback on at the final design review at the end of sprint 3.

The manufacturing phase also took much longer than expected, since both members of our group needed to get machine shop training, and we had to resolve some weight and budget issues before we could proceed with that phase of the process.

Throughout the entire design process for California, here are some key takeaways that we have for new members of CRC. Firstly, familiarize yourself with the machine shop early. Experiment with not only the tools that we have in the ELL, but also go into the machine shop and experiment with lathes and mills before working on parts on your bot. Another thing is to practice soldering. Prioritizing early hands-on soldering training will significantly improve the assembly process in building the bots.

3 California Roll Overview

3.1 California Roll Overview

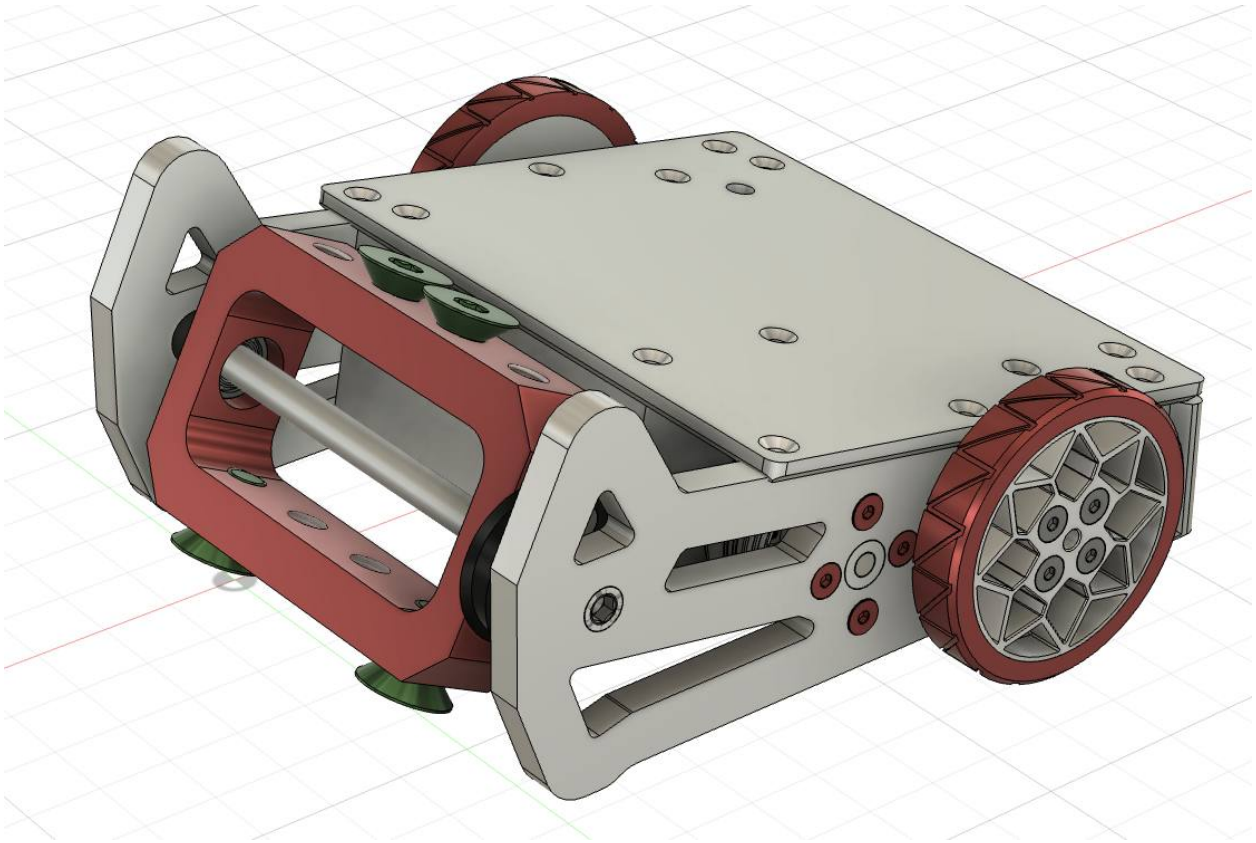


Fig. 3.1.1 California Roll

California Roll is one of the five robots in Infinity's Spring 2026 Robot. Some notable features of the California Roll are its toothed vertical eggbeater spinner, two-wheel skid-steer drive with two directly-driven back cleated wheels, back wedge, and an asymmetrical chassis.

3.2 California Roll Subsystem Overview

Weight (Total: 0.984 lb)	California Roll	Weight Percentage
Weight Chassis	0.3 lbs	30.49%
Weight Drivetrain	0.181 lbs	18.43%
Weight Electronics	0.2 lbs	20.34%
Weight Weapon	0.303 lbs	30.75%

Chart 3.2.1: California Roll Weight Breakdown, by Subsystem

Overseeing California Roll's weight distributions across its varying subsystems, it is evident that the weapon and chassis are the heaviest parts of the entire battlebot. We intentionally designed our 1lb bot to follow the proportions of a traditional 12lb vertical beater bar robot, allocating roughly 30% of our total mass each to the chassis and weapon systems to maximize our kinetic energy footprint.

3.3 Decision-Making Process

Our decision-making process was initialized by redefining our subteam goals, and then brainstorming a variety of different types of antweight robots with varying weapons in order to build one successfully.

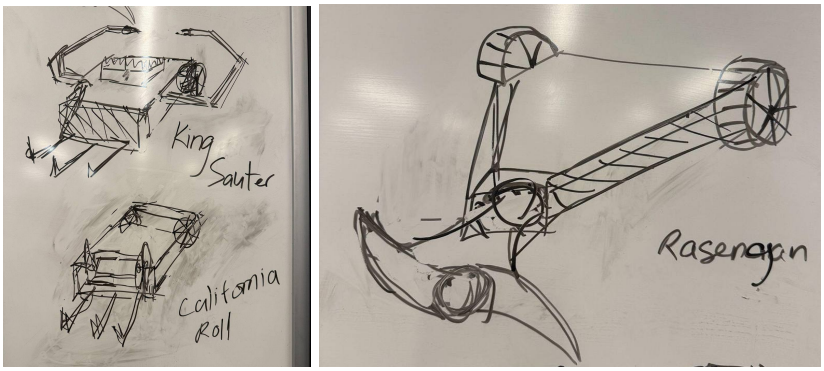


Fig. 3.3.1 Pugh Decision Matrix

A notable difference from last year was our break from the previous system of doing Crazy-8s. Previously, Crazy-8s were our first step in ideation to get a general idea of a robot's main characteristics and to incorporate multiple components from different designs. However, as a subteam, we had very little team knowledge of what makes a kinetic robot successful, which impeded ideation, and we decided to focus on implementing and improving successful robot designs over the years, rather than creating an out-of-the-box, Frankensteinized robot that juggles multiple new CRC and NHRL concepts.

- Manufacturability: components that can be manufactured in-house (with mills and lathes in the Machine Shop)
- Competitiveness: a robot that is successful in the current meta.
- Affordability: subsystems that fit within our $500 budget.

With these criteria in mind, we used a Pugh Decision Matrix to determine which of the three ideas that we had in mind would be bot that could serve our needs:

Criteria	Weight	Rasengan	King Sauter	California Roll
Servicability: How easily could we make this easy to service or repair mid-competition?	6	1	-1	1
Drivability: Is a moderately skilled driver able to control the robot with no problems? i.e. Easy to drive precisely	7	1	0	1
Personal enjoyment: Vibes only - depends on the person. Fill this out only after considering all other categories!	6	0	1	1
Invertability: How easy is it to implenent an invertible aspect? How easy is the process of uninverting?	7	1	-1	1
Creativeness: How different is it from existing NHRL and CRC robot designs?	4	-1	1	-1
Manufacturability: How doable is this design given our current manufacturing constraints?	8	1	0	1
Simplicity (Is it easy to make?)	7	1	-1	1
Competitive (Is it likely to win?)	9	1	0	1
Affordability (Do we have the money for this?)	9	1	1	1
Originality?	5	0	1	0
Durability (survive drop fall? multiple hits?)	7	1	-1	1
Damage/Aggression (do you intimidate other people and pack a punch?)	8	1	1	1
Assembly time (does it take a long time to put together?	7	0	-1	0
Design flexibility (how much can we iterate, customize, and improve?)	5	1	1	1
Aesthetic Potential (Can we make it cool/intimidating)	10	0	1	0
	Totals	69	3	75
	Rank	2	5	1

Fig. 3.3.2 Pugh Decision Matrix



Fig. 3.3.3 3 Bot Criteria Matrix

Although King Sauter looked the most interesting to us, due to its unique weapon assembly using soldering irons, that battlebot did not suffice for our requirements. We decided on a more reliable weapon assembly that would be easier to replicate from previous bots and help from oldies in the Kinetic and Sportsman subteams. Thus, we decided on a vertical spinner eggbeater: California Roll.

4 Weapon

4.1 Toothed Vertical Eggbeater Spinner

One of the primary goals for California Roll's weapon design was achieving a strong balance between reliability, bite, and compact packaging. Thus, we selected a symmetrical aluminum drum with hardened steel teeth. Compared to more aggressive single-tooth or asymmetrical designs, a two-tooth setup provided a better compromise between bite, stability, and tuning simplicity while still allowing strong impact performance. The drum body was designed primarily from 6061-T6 aluminum to reduce weight and simplify machining, while hardened steel teeth were used to maximize durability and energy transfer during impacts.

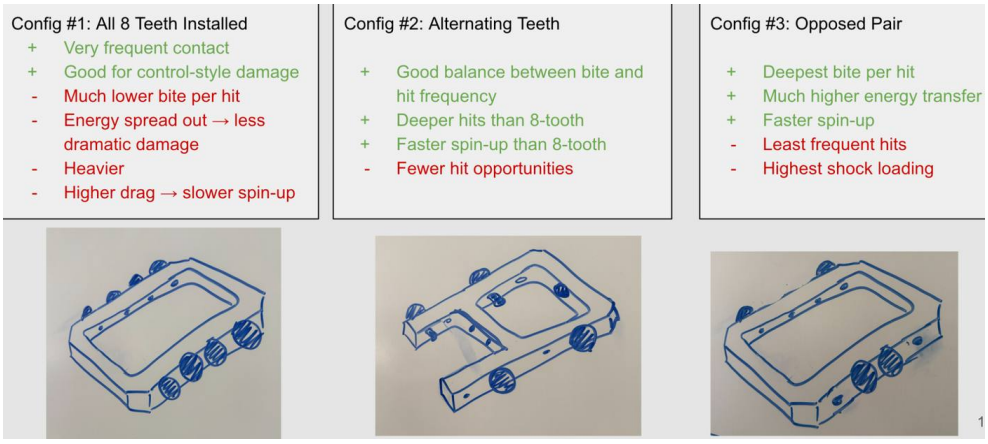


Fig. 4.1.1 Different Eggbeater Teeth Configurations

During the early design process, our team explored several different tooth configurations to balance bite, hit frequency, and overall energy transfer. Higher tooth-count configurations would provide more frequent impacts but lower bite per hit and slower spin-up due to increased weight and drag. Lower tooth-count configurations allowed deeper engagement, larger energy transfer, and faster spin-up, at the cost of less frequent contact and greater shock loading. After evaluating these tradeoffs, we selected a four-tooth opposed configuration (Config #3 in the figure above) as the best balance between destructive capability, stability, and tuning simplicity.

Moreover, the weapon body was designed primarily from 6061-T6 aluminum to reduce weight and simplify machining, while hardened steel teeth were chosen to maximize durability and impact effectiveness. The drum geometry was intentionally kept as a mostly 2D-profiled part so it could be manufactured in-house more easily. The tooth mounting holes were designed to be tapped directly into the weapon body, allowing the hardened steel teeth to be attached.

The final geometry is as follows:

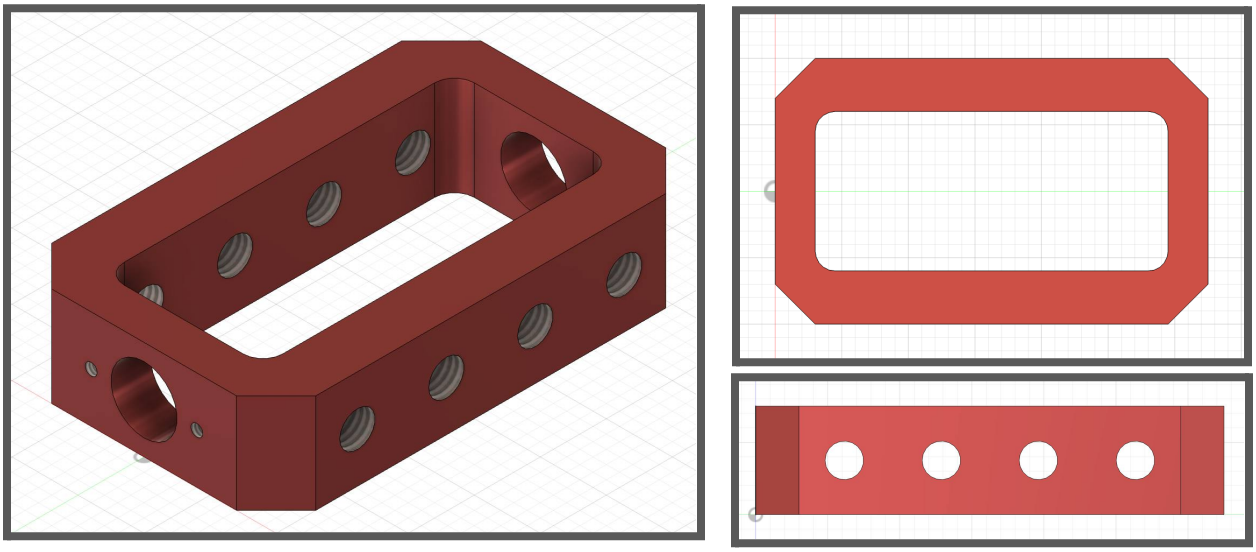


Fig. 4.1.2: Eggbeater Diagrams

With our geometry, we were able to attain the following Eggbeater properties (6061 Aluminum):

Weight (lb)	0.310
MOI (lb•in^2)	0.11925

4.2 Weapon Assembly

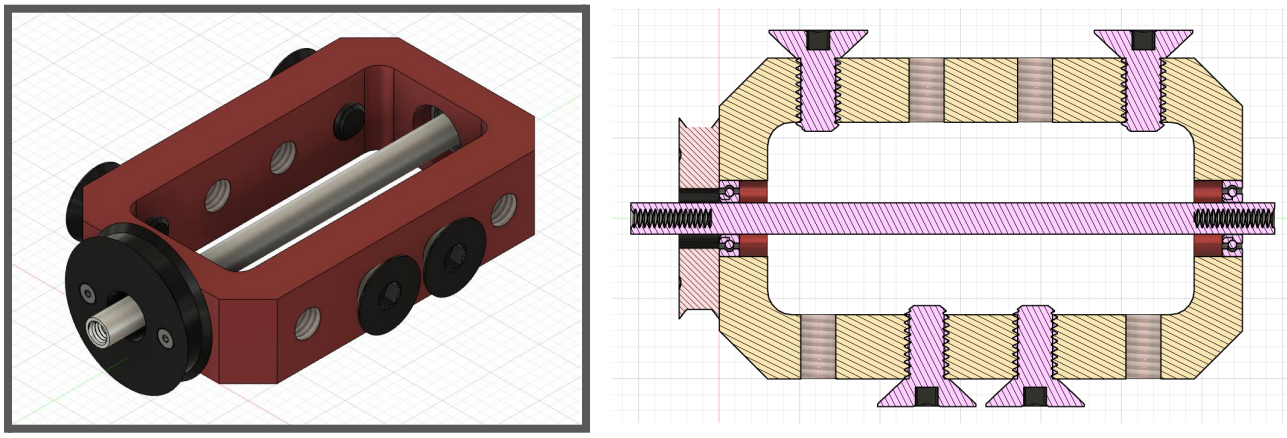


Fig. 4.2.1: Weapon Assembly and Cross Section

California Roll's weapon assembly is built from a few key components: the weapon body, the hardened steel teeth, a TPU pulley, an aluminium dead shaft, and two Nylon spacers.

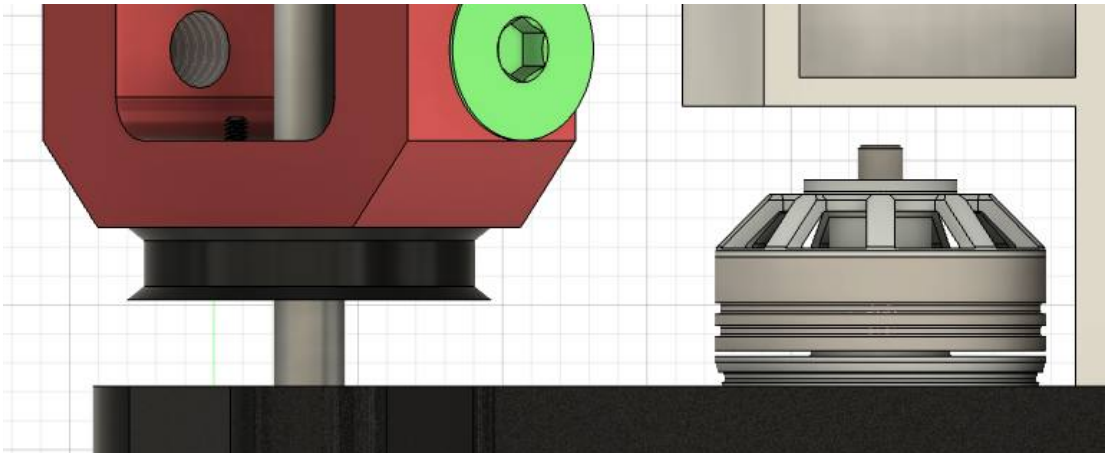


Fig. 4.2.2: Weapon Actuation

The assembly is driven directly from the outer face of an outrunner motor through a timing belt and pulley system, a configuration structurally similar to the actuation method used in *Jormungandr Sportsman's 2026 Spring robot*, where torque is transferred from the motor bell via a belt drive to the weapon pulley. This approach improves packaging efficiency while maintaining high torque transmission fidelity and reducing shock loading on the motor bearings.

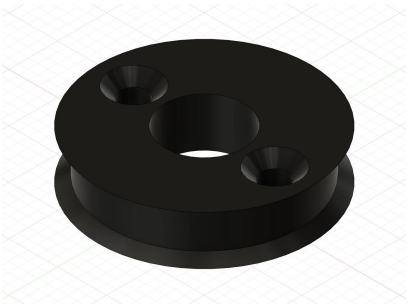


Fig. 4.2.3: Weapon Pulley

The weapon pulley system uses a 1.00:1 pulley ratio, selected based on the updated weapon specification model to optimize spin-up efficiency and current draw. Under this configuration, the system achieves approximately 21,285 RPM within 2 seconds, a tip speed of ~149 mph, and a 2-second kinetic energy of 87 J. Spin-up performance reaches 95% in approximately 0.57 seconds, indicating a highly responsive drive system for short-duration engagements.

4.3 Weapon Speccing

The weapon configuration process for California Roll was driven by an iterative parameter sweep across motor KV, pulley ratio, and current limits to optimize the trade-off between spin-up time, stored kinetic energy, and electrical efficiency. This was evaluated using a simulation-based weapon model, where each configuration (CALI 1–CALI 4) was assessed under standardized 2-second performance metrics to capture both acceleration behavior and steady-state limitations under load.

Weapon Config Name	Max Current [A]	2-second Kinetic Energy [J]	2-second RPM	2-second Tip speed [mph]	95% Spinup Time [s]	Total Kinetic Energy [J]	Tip speed [mph]	Battery Used [mAh]	Continuous Current [A]	Angular Momentum	Gyro Precession Rate [rev/s]
CALI 1	24	25	11541	81	0.38	25	81	2	0	0.042	1.17
CALI 2	28	47	15737	110	0.49	47	110	3	1	0.058	0.86
CALI 3	40	87	21285	149	0.57	87	149	6	3	0.078	0.64
CALI 4	64	139	26995	188	0.57	139	188	10	6	0.099	0.50

Fig 4.3.1 Weapon Speccer Sheet

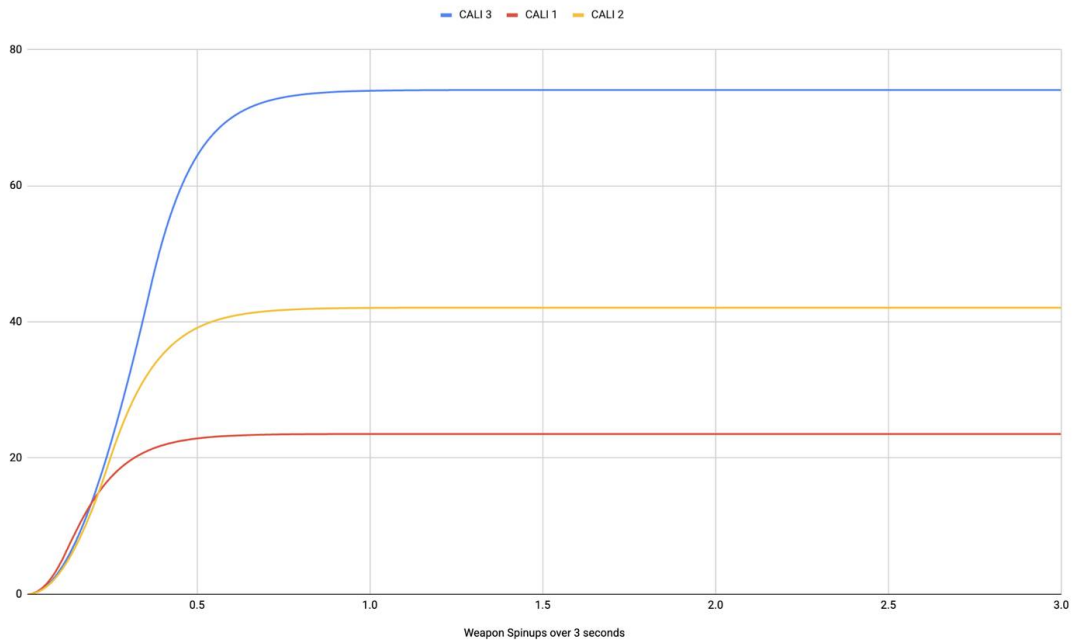


Fig 4.3.2 Weapon Spinup Graphs

Initially, CALI 4 (2500 KV) was the intended baseline design due to its highest theoretical performance, offering the greatest peak RPM and kinetic energy output within the tested configurations. However, this option was ultimately not viable due to supply constraints on the selected weapon motor. As a result, the design was re-optimized around available hardware, leading to the final selection, CALI 3 (1960 KV), which represents the best achievable balance between performance and availability.

4.4 Machining

The process of manufacturing the weapons involved facing and cutting ¾" x 2" 6061 Aluminum bar stock to size using a bandsaw. After that, a slot with filleted corners would be cut on a manual mill using a ⅛" endmill after first drilling a large enough hole to put the endmill in the middle of the material. To achieve the correct dimensions of the slot, rectangular passes were performed, gradually increasing the width and height of the slot. After the basic shape of the weapon was established, the M8 tooth holes and M3 pulley attachment holes were drilled and tapped. Finally, the corners of the weapon were chamfered using a 90° bit.

For the UHMW uprights, we removed thickness by standing them up in the mill vise and taking material off in horizontal passes. Then, we 3D-printed a custom PLA jig for the uprights to drill the holes in their faces, since the upright geometry did not enable us to clamp it in the vise directly.

5 Electronics

5.1 Weapon Motor

California Roll uses the BadAss 2305-1960Kv Brushless Motor. This motor achieves approximately 21,285 RPM, 87 J of kinetic energy, and ~149 mph tip speed at 2 seconds, with a 95% spin-up time of 0.57 seconds and a manageable continuous current draw of ~20 A.



Fig 5.1.1: BadAss 2305-1960Kv Weapon Brushless Motor

5.2 Battery

Our two primary considerations when selecting a battery were ensuring sufficient capacity to support the high-energy weapon system and verifying that the pack could handle the required burst current during peak load events. Based on discussions within the NHRL Discord community regarding typical 1 lb-class power systems, we determined that ~300–500 mAh 3S LiPo packs. Based on these constraints, we selected the **Gaoneng GNB 11.1V 3S 450mAh 80C LiPo Micro Battery (Long Type)** with XT30 connector.

During bench testing, we conducted some circuitry testing to monitor heat generation and voltage sag under peak load. A major issue arose when we connected the hot and live wires to the wrong ports in the XT30 connector. This immediately caused a LiPo fire inside the ELL, leading us to only have one working battery during the May competition.



Fig 5.2.1: Gaoneng GNB 11.1V 3S 450mAh 80C LiPo Micro Battery

5.3 ESCs

For the weapon ESC, we selected the **Readytosky 35A BLHeli_S Ant Weapon ESC**. It is a compact 2–6S ESC with a 35A rating and pre-tuned BLHeli_S firmware designed specifically for anti-weight weapon systems. Its low mass and proven reliability in high-RPM brushless weapon applications made it a straightforward choice for our design. For the drive, we selected the **Repeat Micro AM32 Drive ESC**. This ESC was chosen for its extremely small form factor (12 × 18 mm), ~5 g weight, and AM32 firmware tuned for smooth bidirectional brushless control.

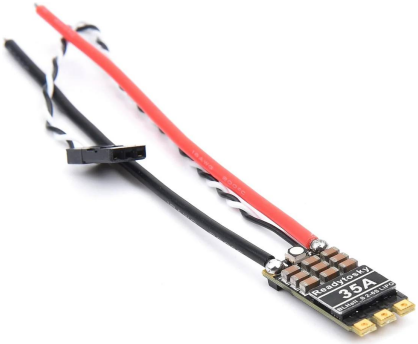


Fig 5.3.1: Readytosky 35A blheli_s Ant Weapon ESC

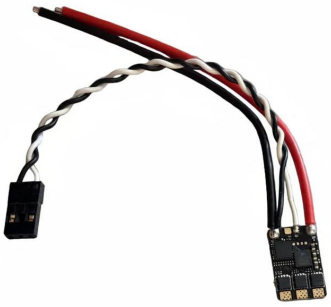


Fig 5.3.2: AM32 Micro Drive ESC

5.4 Drive Motor and Gearbox

Each of the 5 antweights bots on the Infinity subteam utilizes the same drive motors and gearboxes. This was done for simplicity and efficiency purposes. The drivetrain integration was one of the most difficult parts of the overall assembly process. We first had to fully disassemble the **Repeat Robotics Mini Brushed Mk2 Antweight planetary gearmotors** to swap in the correct internal motor configuration, which was complicated by a lack of properly sized precision screwdrivers. After reassembly, we also ran into reliability issues where the motor lead wires repeatedly detached from the hub. This required us to resolder all four motor pairs, making the process especially delicate and time-consuming.



Fig. 5.4.1: Mini Brushed Mk2



Fig. 5.4.2: Mini 1106 Mk4: planetary gearmotor

5.5 Transmitter & Receiver

We decided to use a transmitter that has been previously used a lot in past Sportsman and Kinetic battlebots. We figured that the **Flysky FS-i6X 6-10 (Default 6) CH 2.4GHz AFHDS** was a good option. We also decided not to use the stock receiver due to weight-saving reasons, so we used a smaller receiver: Flysky FS2A Receiver.

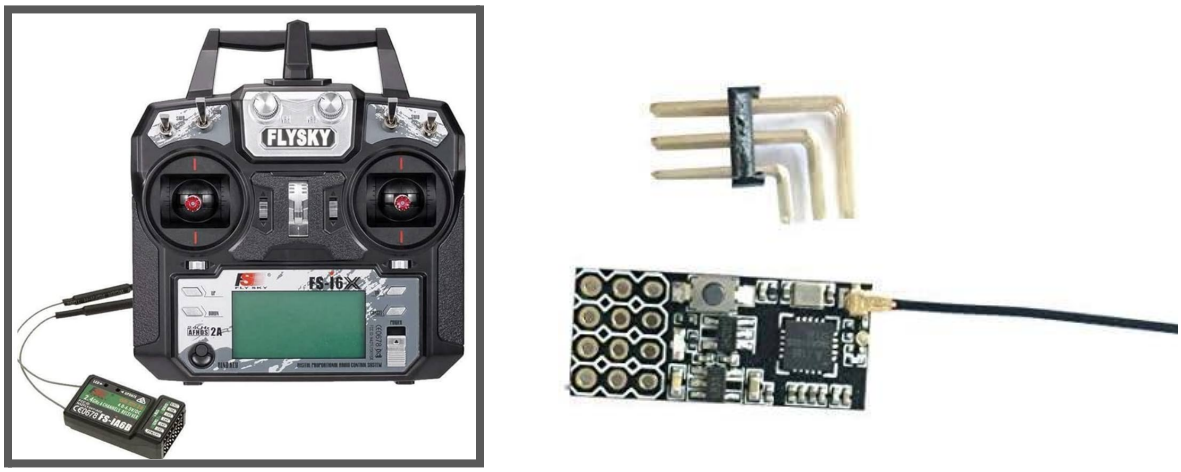


Fig 5.5.1 Transmitter and Receiver

5.6 Circuit Diagram

Unlike the standard Sportsman and Kinetic circuit diagrams, we decided to directly solder our drive and weapon motors to our drive and weapon ESCs. This way, we could reduce the amount of weight by preventing the use of XT30 connectors.

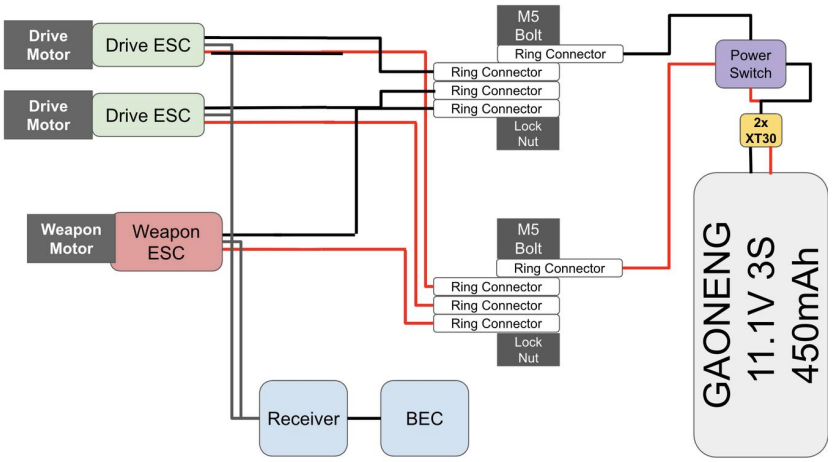


Fig 5.5. California Roll Circuit Diagram

6 Chassis

6.1 Billet

Our billet was designed after our uprights, using the sketch we had previously created including the geometry of all relevant electronics and approximating thicknesses for and the integrated bottom layer. These thicknesses were iterated after slicing our plate to ensure we remained within weight while providing adequate protection for internal components in conversation with experienced antweight builders from the NHRL discord. We decided to use TPU for the billet based on other 1lb bots we took inspiration from and based on the TPU billets of the 12lb bots being built this spring, which use the material due to its low mass and high compliance, which helps absorb impacts and protect critical components. The internal walls of the billet were designed to protect the wires running through the middle of it from damage from the spinning drive motors, and everything was filleted to improve stability. At the last minute, we added countersunk holes to potentially mount the drive motors to, when we weren't certain the screws of the correct length would arrive on time, and we had minimal print time at our disposal.

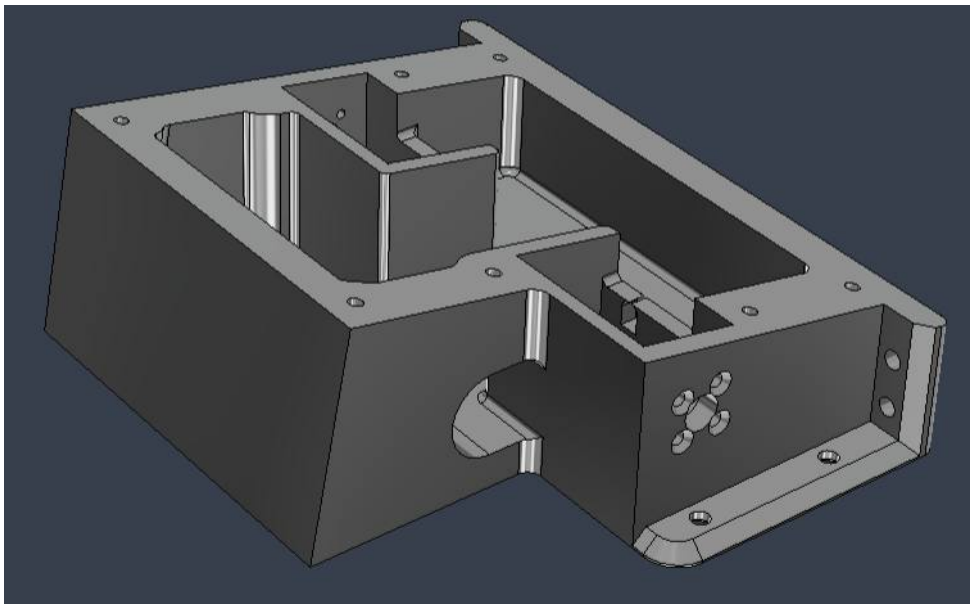


Figure 6.1: Billet

6.2 Uprights

The two mostly symmetrical uprights are fabricated from 0.25-in (6.35 mm) ultra-high molecular weight (UHMW) polyethylene (machined down from 0.375-in, 9.525mm) to provide stiffness and serve as the primary load paths for the TPU chassis, the drivetrain, and the weapon assembly. The left upright also serves as the mounting surface for the weapon motor.

Designing the uprights involved a delicate balance between ensuring a center of mass as close to the drive axle as possible while ensuring the billet (which was designed in tandem with the uprights) has enough space to accommodate the electronics. We ensured that the bot would be invertible by using the inverted ground plane (a line running tangent to the wheel and the tip of the upright), which was unobstructed by the upright, weapon, and billet. Below is the sketch for our left upright, demonstrating how we optimized its form according to both the parts it needed to support while preserving invertibility.

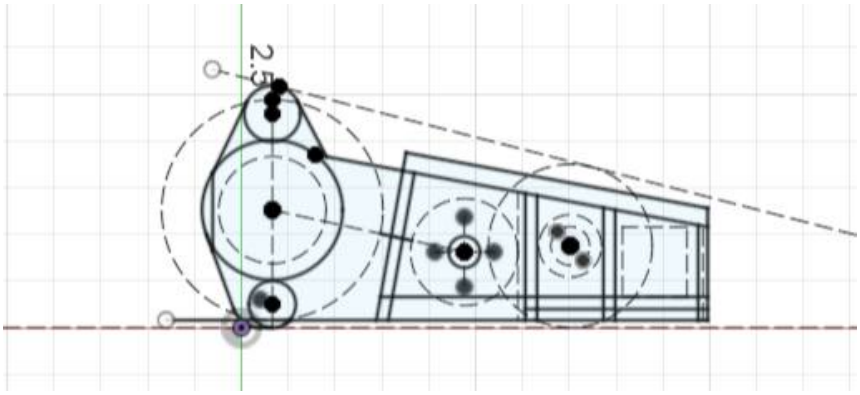


Fig. 6.2 - Upright sketch with inverted ground plane

We also carefully designed weight-saving holes in our uprights to remain within the 1lb weight constraint. These holes were necessary due to the use of UHMW rather than a lighter printed material, and required optimizing holes to ensure enough material remained around all screw holes and especially around the weapon, where we suspected the greatest amount of impact would be sustained. Additionally, we ran simulations to confirm the new design could sustain the forces generated by the impact of another 1lb vertical spinner (using team NoBull's weapon CAD to simulate an opponent attack).

Our initial design, with weight-saving holes, required additional modifications to meet SendCutSend manufacturing specifications, so we iterated on our previous design to ensure they would be able to manufacture it. Specifically, we had issues with the minimum hole size, minimum bridge, and minimum hole to edge distances. Attached is an image of these specifications for 0.375" UHMW and the problematic parts of the old upright design. Avoiding similar issues in the future will enable us to minimize lead times on SendCutSend parts by ensuring no follow ups on their part are necessary.

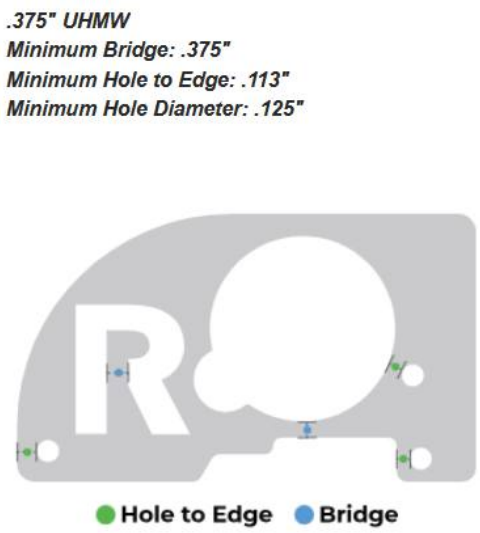


Fig 6.2.1 SendCutSend specifications for .375" UHMW

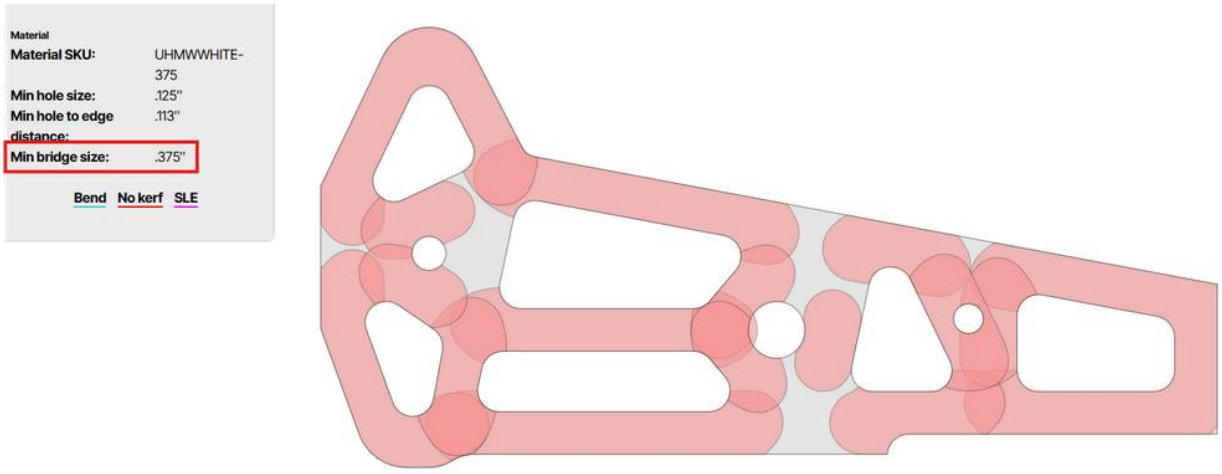


Fig 6.2.2 Issues with manufacturing our original upright design

In this process, we needed to eliminate all the mounting holes (for the M2 and M3 screws attaching the weapon and drive motor) from the design we sent in, which meant we then had to machine all those holes in-house. We also drilled all the pilot holes for the plastite screws attaching the uprights to the top plate and billet ourselves.

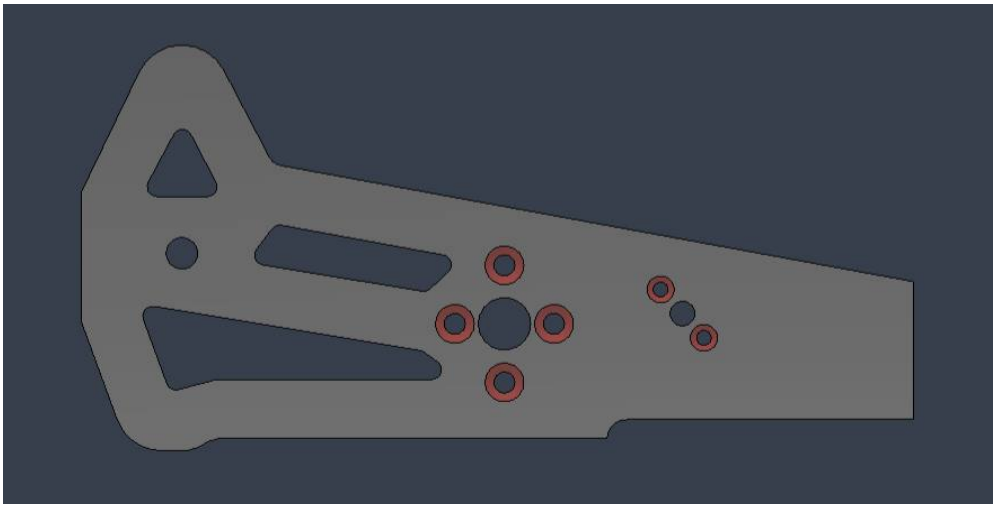


Fig. 6.3 - Upright design with machined holes indicated in red

A key topic of debate in the later phases of our design process was whether the uprights should be 3D-printed using TPU or PA6-CF filament (which most other Infinity groups chose to do) rather than UHMW. The advantage of printed uprights would be not only weight savings, but the potential for iteration while remaining in budget. Our UHMW uprights cost about 10% of our total budget, which could have been spent on spare electronics. Nevertheless, we decided to proceed with the UHMW uprights, not only because we were more certain of their ability to withstand impacts and support our entire weapon and drivetrain systems, but also because we believed it would give us more relevant experience for our future work on 12lb robots.

Because we didn't have enough weight for a dedicated bottom plate made out of a more rigid material, we made the bottom of our billet thicker and extruded a lip to screw plastite screws through and into the bottom of our uprights; visible below is the notch we designed to accommodate that new lip.

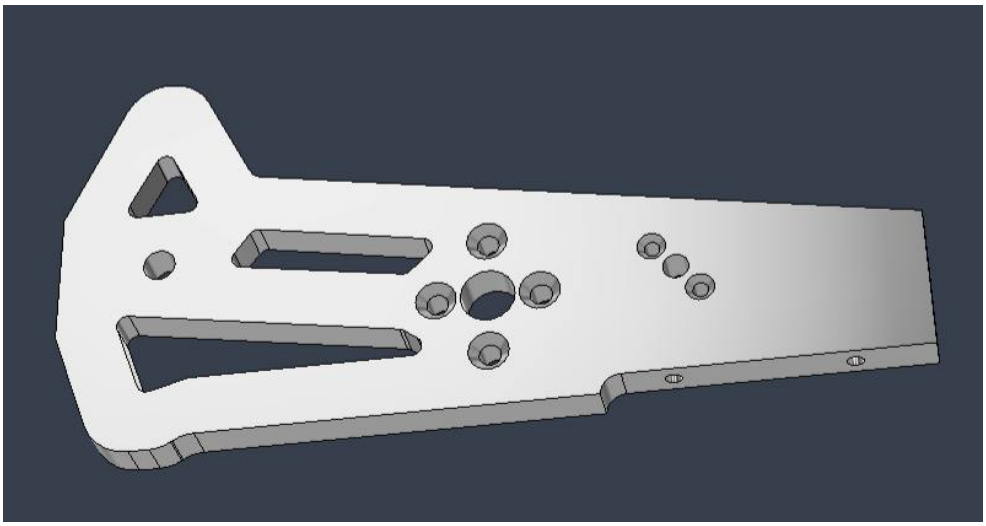


Fig. 6.4 Left Upright

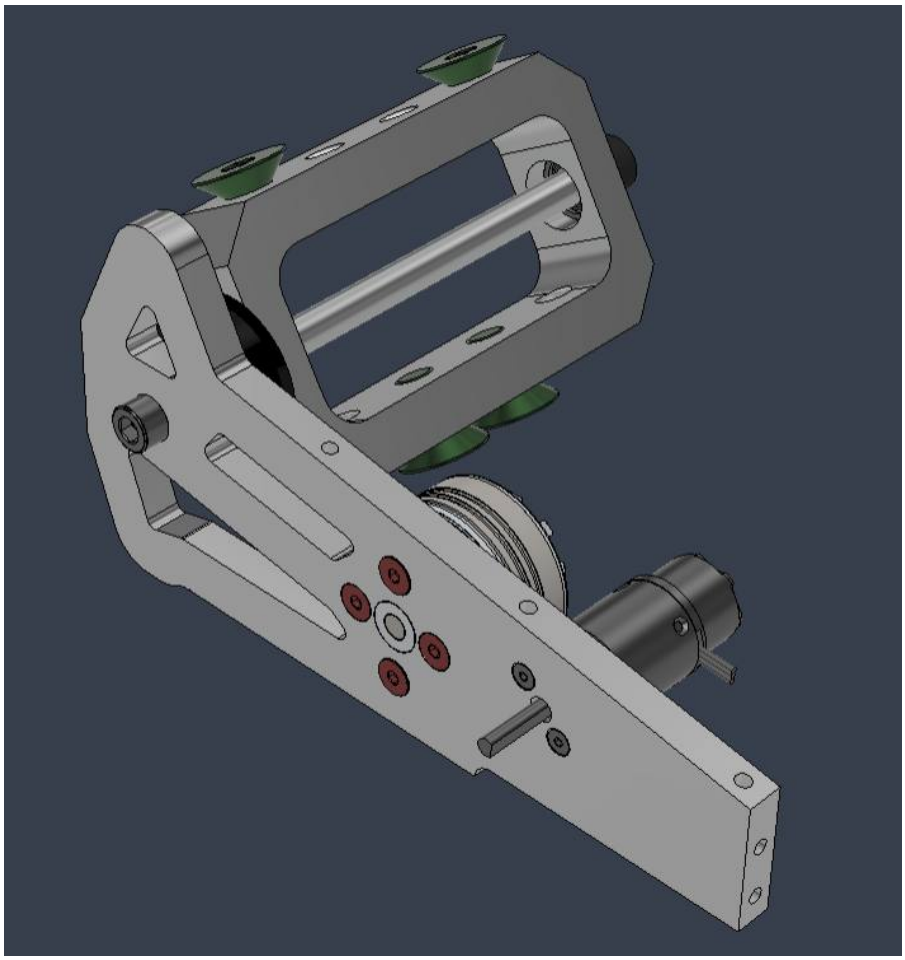


Fig. 6.4.2 - Left upright with weapon, weapon motor, and drive motor mounted

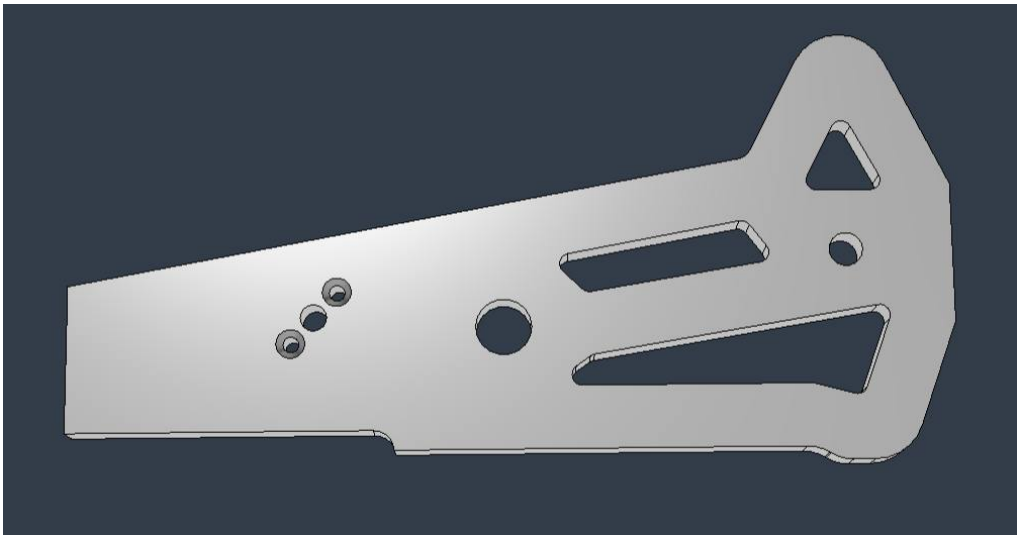


Fig. 6.5 Right upright

6.3

6.4 Top Plate

To enclose the internal electronics while maintaining strict adherence to our 1-pound weight budget, the top plate was 3D-printed from TPU with a thickness of 4 mm using a 3D honeycomb infill pattern. This geometry drastically reduced component mass while providing exceptional compliance, allowing the plate to flex and absorb energy from overhead hits without fracturing.

The top plate and billet lips were secured into the UHMW uprights using **23 Plastite screws**. This significantly ensured our top plate would not fall off during competition, learning from Benny and Capsize's past performances. Reflecting on this after assembly, this volume of fasteners was excessive. Over-engineering the mounting points added unnecessary hardware weight that could have been better utilized elsewhere—such as thickening the bottom chassis wall or investing in a dedicated, rigid bottom plate.

During safety tech inspection at the NHRL competition, we modified the top plate to include a dedicated **power switch hole** to comply with the rule requiring external bot activation pre-fight. Because our initial CAD lacked this feature, we manually cut an access port into the printed plate. This allowed the power link to be easily reached by a tool before the match while keeping the internal wiring and LiPo battery shielded from opposing weapon teeth during combat.

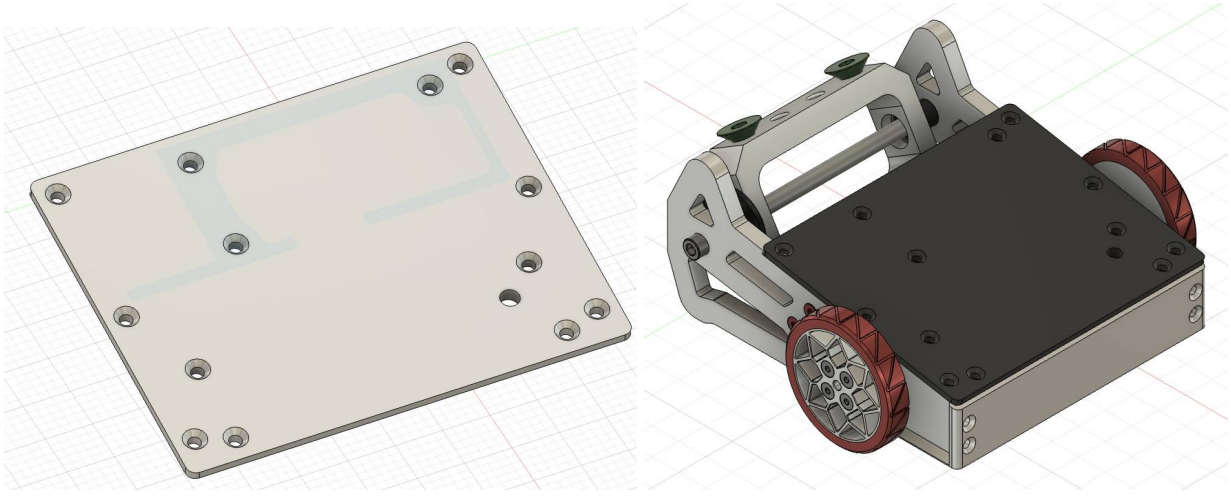


Fig. 6.7.1: TPU Top Plate and Top Plate on

7 Drivetrain

7.1 Actuation

California Roll implemented a 2 Wheel Direct Drive Skid Steer drivetrain assembly. We believed that using this drive assembly rather than implementing a 4WD drivetrain system would be a more suitable option for the antweight category. The primary concern with choosing this system was due to weight constraints, as the extra motors, gearboxes, and belts would have taken valuable weight away from our heavy weapon assembly.

Moreover, we decided to implement a direct drive system not just because of weight constraints but because it would have significantly simplified the packaging of wires and motors of our overall bot, whilst simultaneously reducing the number of total failure points across the chassis.

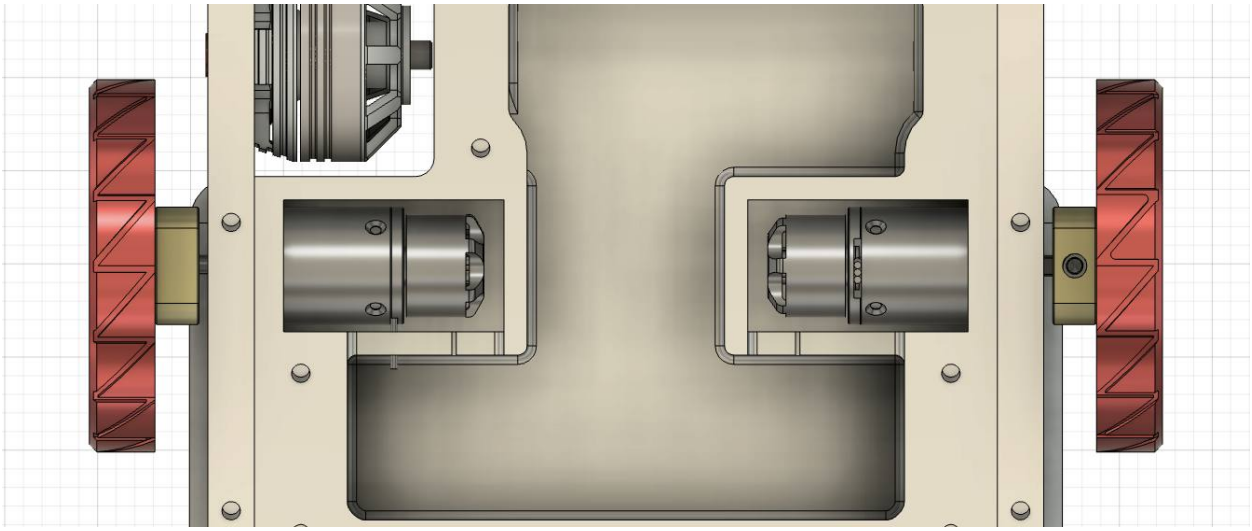


Fig 7.1 Drive Assembly

7.2 Wheel Guards

Wheelguards were a primary addition to California Roll. However, due to weight constraints, we decided to remove them. Nevertheless, below presents the ideation that we had behind these interestingly shaped wheelguards.

Initially, our idea on how the wheelguards would be shaped was inspired by “Knuckle Sandwich”, advised by CRC oldie, Zander Knight. We thought that because our drivetrain assembly was supported by a 2-wheel skid steer system, our wheels were a very vulnerable component that, if hit, would severely impact our bot's performance.



Fig 7.2.1 Knuckle Sandwich

Mimicking this design, we decided that using TPU was the most optimal material for wheelguards: this geometry allows California Roll to take advantage of TPU's natural flexibility, allowing the guards to bend and absorb massive hits from opposing weapons instead of passing those destructive forces straight into our delicate drive motor gearboxes. Our wheelguards are also aligned with the slantedness of the billet, allowing for a seamless integration between our uprights, wheelguards, and billet.

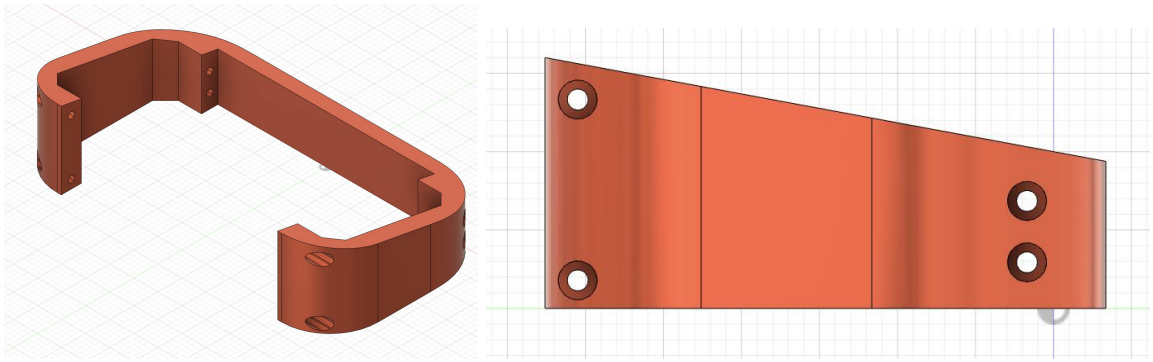


Fig 7.2.2 California Roll’s Wheelguards

7.3 Wheels

The primary inspiration for our wheelbase came from the Forza Performance 21 Inch rims for the BMW M8 Competition. During the Infinity showcase, one primary problem with getting enough traction emerged: the wheels seemed to barely get any contact with the ground, and even during the times that they did contact the ground, they overspun a lot, rather than gripping the floor. This caused significant issues with not only our drive across the cage, but also with our weapon being able to get a strong bite into our opponents.

During the revamp process for competition, we opted for a larger wheelbase diameter of 2 inches, a polyurethane thickness of 0.354 in. This change in wheelbase dimensions significantly improved our drive, improving our agility and overall maneuverability.



Fig 7.3.1 BMW M8 Forza Performance Rims vs. California Roll Rear Wheel Assembly

Another improvement we made was increasing the depth of our tread embossing. By making the tread pattern deeper into the polyurethane, the wheels were able to dig much better into the arena floor texture instead of just sliding across it. Additionally, California Roll's wheel treads were inspired by Zander Knight's Cicero. We also introduced pocketed cutouts that closely mocked the sleek geometric look of the BMW rims, which successfully removed heavy material from the thickest parts of the wheel hub where it wasn't structurally needed.

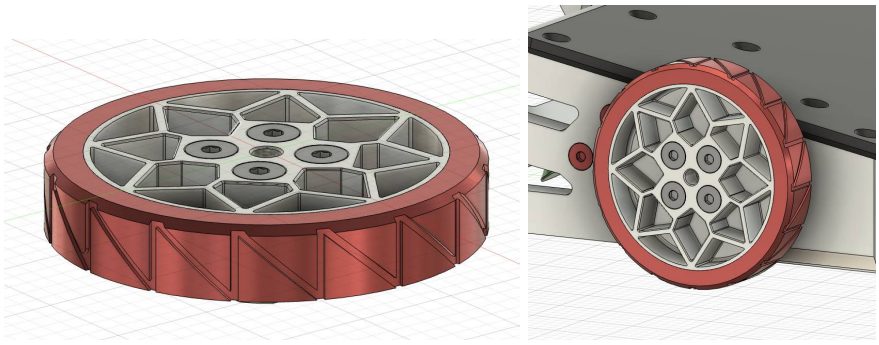


Fig 7.3.2 Wheel Side Profiles

8 Next Steps

Since the CRC Infinity showcase, we believed that our primary concern with our weapon actuation was actually our drive: we couldn't get that deep of a bite because we couldn't drive into our opponents due to the gyroscopic force of our weapon. Acknowledging this mishap, we opted for a larger wheelbase: not only a larger radius, but an increased thickness for our wheels, to improve our overall traction.

California Roll had a strong start during the May 2026 NHRL. The improved drivetrain assembly significantly improved the traction between the wheels and the ground of the cage, drastically improving Cali's overall agility and speed. Not only did this help us maneuver around the cage more strategically, but it also allowed us to generate stronger drives into our opponents, allowing for an overall stronger hit.

Despite this increased hit speed and magnitude, we found that one of our biggest challenges at NHRL was getting enough of a bite against our opponents. The dimensions of Cali's weapon, combined with the positioning of the weapon assembly in the uprights, did not allow Cali to have deep bites into the opponents. With Cali's high weapon spin rate, the weapon would chip off the opponents along the edge before going deeper.

The improved drivetrain setup significantly helped us with our drive and weapon hitting throughout NHRL. But we realized that for future ideas, we should revamp the structure of our weapon assembly so that our weapon could reach further out into the opponent, thus allowing for a larger bite.

9 Acknowledgements

Thank you, Ethan Matthew Pua Zhang, and James LaSalle, for putting so much effort into creating and organizing Project Infinity for us. We genuinely have learned so much this past year; not only that, we had so much fun making our own 1lb projects.

Also, a special thanks to many of the CRC oldies who helped us out in this process, whether it was soldering, CADding, or ideating and helping us alternate our design decisions. We could not have done this without your help.